ONBOARDING DOCUMENT · INTERNAL USE

Intern Task Allocation

Upwork SEO Bot & Listing Website — module assignments, hardened deliverables, and review criteria for the one-week trial.

⏱ Deadline: 1 Week from First Meeting

Resources & Reference Links

Loom Videos (Watch All Before Meeting)

VIDEO 1 **Project Overview & Bot Demo**
https://www.loom.com/share/b0c41f5c641a4a54a5d5c7eb610f54e0

VIDEO 2 **Database & Flow Walkthrough**
https://www.loom.com/share/93609e9813bc45c7bf7269bc7e8e722a

VIDEO 3 **Cron Job & Integration Demo**
https://www.loom.com/share/0cc13315cf7c4761bcf2378724dbe9b6

Reference — Upwork Search Query

https://www.upwork.com/nx/search/jobs/?q=(apify OR automation OR make.com OR n8n OR cold OR calling)&sort=recency

Project Overview — Existing Project

This section explains what the project currently does. You are being given an existing codebase and your task is to build on top of it. Read this carefully to understand what is already in place and what still needs to be done.

The project has two main parts:

1. Upwork Electron App (JavaScript)

• This is the frontend application built as an Electron desktop app.

• Campaigns are created and managed from this app.

• Inside it, an Upwork Discord Bot helps fetch job listings from Upwork.

• When a campaign runs, it first performs job listing — scraping jobs from Upwork based on the search query.

2. Upwork SEO Board (Python)

• This is the backend that processes the scraped jobs.

• After jobs are listed, a viability prompt is sent to ChatGPT to evaluate each job (Yes / No).

• If viability = Yes, a GitHub repository is created for that job — this is what we call the Product Page.

• The project currently uses MongoDB as its database (migration to PostgreSQL is part of your task).

What is NOT yet implemented (your tasks):

- Blog Page generation — needs to be built from scratch
- Service Page generation — needs to be built from scratch
- PostgreSQL migration — currently on MongoDB, must be moved to PostgreSQL
- Listing Website — a separate website to display all content (jobs, products, blogs, services)

⚠ **Known Bug:** The ChatGPT scraping is currently not working correctly. The DOM scraping logic may be broken or outdated. Fixing this is part of your task.

Module 1 — Upwork SEO Bot (To-Do List)

#	Task	Details	Done
1	**Remove GitHub Repository Functionality**	Remove the GitHub repository making functionality entirely from the existing codebase.	☐
2	**Remove Proxies**	Remove all proxy configurations and related code from the project.	☐

#	Task	Details	Done
3	**Migrate Database: MongoDB to PostgreSQL**	Move the entire database from MongoDB to PostgreSQL. Ensure all existing data models and queries are updated. **Corruption & Recovery Proof (Required):** After completing the migration, intentionally drop one column from the `jobs_selected` table while a campaign is actively running. Capture the exact error output (with timestamp and line number) in a file called `failure_log.txt`. Identify the exact line in the codebase where the error was thrown — not where it was caught, where it was thrown. Fix the schema and include a before/after screenshot of the table in a PostgreSQL client (TablePlus, pgAdmin, or psql). Submit `failure_log.txt` and both screenshots with your final deliverables. **⚠ Non-negotiable — this proves the migration is wired into the live pipeline and not just a schema dump.**	☐
4	**Fix ChatGPT Scraping Issue**	Resolve the current issue with ChatGPT DOM scraping. Before writing any fix, you must first submit a **Diagnosis Report** as a separate file called `diagnosis.md` containing: (1) a screenshot of the actual Puppeteer browser showing the broken DOM, (2) the exact failing selector from the code and a clear explanation of why it no longer matches, and (3) 3 sentences in your own words explaining the root cause. Your code fix must include a comment block referencing your diagnosis by selector name. In the second meeting, you will be asked to show the broken state and the fixed state back-to-back. **⚠ A fix submitted without the `diagnosis.md` file will be treated as incomplete.**	☐
5	**Create 4 Database Tables**	Create the following PostgreSQL tables: • `jobs_selected` — Jobs marked as viable (viability = 'Yes') • `product` — Auto-generated product pages • `blog` — Auto-generated blog posts • `services` — Auto-generated service pages	☐
6	**Create 2 Separate Prompts**	Write and integrate 2 separate ChatGPT prompts: • Blog generation prompt • Services generation prompt	☐

Module 2 — Listing Website (To-Do List)

#	Task	Details	Done
1	**Create Listing Website**	Build a separate listing website with its own database (not the same DB as Module 1). This website will display jobs, products, blogs, and services publicly.	☐
2	**Create Cron Job**	Set up a cron job that runs every 15 minutes. It must be implemented with a watermark and audit log — not a naive full-copy. Specifically: (1) store a `last_synced_at` watermark per table in the listing website's own database so the cron never re-copies already-synced rows, (2) every run must insert one row into a `sync_audit_log` table with the fields `run_at`, `table_name`, `rows_synced`, and `error` (null if the run was clean). To prove it works under failure: kill the Module 1 database connection mid-sync, let it fail, then show the resulting `sync_audit_log` row containing the captured error message. Your Loom demo must show a minimum of 5 audit log rows including at least one failure row and one successful recovery row. **⚠ A cron job submitted without a watermark or audit trail will not be accepted.**	☐
3	**Display All Content**	Ensure the website properly displays: • Job listings • Product pages • Blog posts • Service pages Every new project should appear on the listing website automatically.	☐

Deliverables

You must submit **ALL** of the following by the deadline. Incomplete submissions will not be accepted.

DELIVERABLE 1 ☐

Loom Video — Full End-to-End Demo

Record a single Loom video demonstrating the entire project working end-to-end. This must include:

- The Upwork bot scraping jobs and saving to PostgreSQL
- The ChatGPT viability check running correctly
- Product page, blog, and service page being generated for viable jobs
- The listing website displaying all content (jobs, products, blogs, services)
- The cron job syncing new data from Module 1 to the listing website

Walk through each step clearly and explain what is happening as you demo.

DELIVERABLE 2

3 Job Samples — Generated Outputs

Provide 3 complete job samples. For each job, show the following generated outputs:

- The selected job data
- The generated Product page
- The generated Blog post
- The generated Service page

DELIVERABLE 3

Diagnosis Report (`diagnosis.md`)

Submit the ChatGPT scraping diagnosis file **before** your fix is reviewed. Must contain the broken DOM screenshot, the failing selector with explanation, and your written root cause analysis. Your fix will not be reviewed without this file.

DELIVERABLE 4

Migration Failure Log (`failure_log.txt`) + Screenshots

Submit the error log from the intentional column drop during live pipeline execution, plus before/after screenshots of the PostgreSQL table in a DB client.

DELIVERABLE 5

Cron Audit Log Output

Paste or screenshot a minimum of 5 rows from the `sync_audit_log` table in your Loom video, showing at least one failure row and one clean recovery row.

Frequently Asked Questions

Q: Can I use a different tech stack or start from scratch?

No. This is an existing, live project. You must build on top of the current codebase. The Electron app is in JavaScript and the Upwork SEO Board is in Python — you will work within these. Do not introduce a new tech stack or rewrite things unnecessarily.

Q: Can I keep using MongoDB?

No. The project currently uses MongoDB but it must be migrated to PostgreSQL. This is a core part of your task.

Q: What does the 'viability check' mean?

The bot sends each scraped job to ChatGPT with a viability prompt. ChatGPT evaluates whether the job is a good fit and responds with 'Yes' or 'No'. If 'Yes', the job is marked as viable and triggers content generation (product, blog, service pages).

Q: How should the cron job work exactly?

The cron job runs every 15 minutes on the listing website's server. It queries the Module 1 PostgreSQL database for any new entries in the product, blog, and services tables and copies them into the listing website's own separate database.

Q: Does the listing website use the same database as the bot?

No. The listing website has its own separate database. The cron job syncs data FROM the bot's database TO the website's database every 15 minutes.

Q: What if I get stuck?

Re-watch the Loom videos first. If still stuck, document your specific question clearly (what you tried, what error you got, what you expected) and reach out. Do NOT wait until the deadline to ask for help.

Q: Do I need to deploy the website?

Not required. A working local demo is sufficient. However, if you finish early and have time, go ahead and deploy it. Show everything working in your Loom video recording.

Q: Can I ask questions after the first meeting?

Yes. There will be two meetings total. The first meeting is for understanding the project and getting your questions answered. After that, you will have ongoing access to ask follow-up questions — but make them specific and clear.

Q: This project seems big. Should I work on other parts of it too?

Absolutely not. This project has many other modules and features beyond what is assigned to you. Do NOT touch, improve, or modify anything outside of the two modules assigned in this document. Do not explore other parts of the codebase. Your scope is ONLY the live Upwork query module and the listing website as described in the Loom videos. Stay strictly within scope.

Q: What if I notice something that could be improved outside my task?

Note it down if you want, but do NOT make any changes to it. Your job is to deliver the assigned modules only. We will discuss improvements separately if needed.

Q: Why do I have to break things on purpose?

Because anyone can write code that works in a clean environment. You are being evaluated on whether you understand the system well enough to debug it when it breaks — which is what real engineering looks like. The failure scenarios are designed so that only someone who actually ran the code will get the right answer.

Q: What counts as the "exact line where the error was thrown"?

The line in your own codebase (not inside a library or Node internals) that triggered the failure. If the error bubbles up through multiple catch blocks, trace it back to the original source call. Paste the file name, line number, and the actual line of code.

Q: Can I record the audit log output separately from the main Loom video?

No. It must appear in the same Loom video as the live demo, shown in a DB client or terminal with visible timestamps. Screenshots submitted separately will not count.

First Meeting Agenda

Before any work begins, you will have an onboarding meeting. The purpose of this meeting is to confirm that you understand the project. **The 1-week deadline starts from this meeting.**

Before the Meeting (Your Homework):

- Watch ALL Loom videos listed in the Resources section — take detailed notes
- Read this entire document thoroughly
- Prepare a written list of questions (anything unclear, anything you want to confirm)
- Spend 1 day understanding the existing project and codebase before the meeting

Meeting Flow (What Will Happen)

Important: In this meeting, YOU will be doing the explaining — not me. I will listen, ask questions, and correct you where needed. **Come prepared.**

1. **You Explain the Project**
   I will ask you: *What is this project? What problem does it solve?* You will explain the project in your own words based on the videos and this document. I will listen and correct you where needed.

2 **You Explain the Flow**

I will ask you: *How does this project work? What is the flow?* You will walk me through the entire flow — from job scraping to content generation to the listing website. I will correct you as you go.

3 **You Explain the Task**

I will ask you: *What exactly do you need to do?* You will explain the to-do list, deliverables, and what the final output should look like. I will confirm or correct.

4 **Your Questions**

You will ask all the questions from the list you prepared. No question is too small — ask everything now. This is your chance to get full clarity before starting.

⚠ **SCOPE REMINDER:** You are ONLY working on the modules described in this document. Do not touch, modify, or explore any other part of the project. Stay within the live Upwork query module and the listing website. Nothing else.

Final Meeting Agenda

The final meeting happens at the end of the one-week trial. This is where the work is reviewed live, your understanding is tested under pressure, and your final marking is decided. **Treat it as a defense, not a demo.**

Before the Meeting (Your Homework):

- Have the entire project running locally and ready to launch in under 60 seconds
- Have all 5 deliverables (Loom video, 3 job samples, diagnosis.md, failure_log.txt + screenshots, audit log output) organised in one folder
- Re-read your own code — you will be asked to modify it live
- Prepare your own list of questions for the CEO (see below)

Meeting Flow (What Will Happen)

Important: In this meeting, YOU will run the demo and YOU will defend the work. I will interrupt, probe, and ask you to change things on the spot. Hesitation and "I'll check later" are not acceptable answers.

1 **Run Code End to End**

You will start the project from a cold state and run the full pipeline live — Upwork bot scraping → PostgreSQL save → ChatGPT viability → product/blog/service generation → cron sync → listing website display. No pre-recorded shortcuts.

2 **Check Website Flow**

You will walk through the listing website page by page — jobs, products, blogs, services. Show how a brand-new entry flows from Module 1 into the website within one cron cycle. Be ready to refresh and prove it live.

3 Question and Answer

I will ask targeted questions about your code, your design decisions, your database schema, and your failure handling. If you don't know an answer, say so — don't guess. Lying about your code in this round is worse than admitting you forgot.

4 Modify One Part of the Code (Live)

I will pick one part of the codebase and ask you to make a small change live — for example, change the cron interval, add a new column to the audit log, or adjust a prompt. You should be able to find the file, make the change, and re-run the affected piece within 10 minutes.

5 Your Questions for the CEO

You bring a written list of questions. Use the question bank below as a starting point — add your own based on what came up during the week. This is your last formal chance to ask anything.

Question Bank — Starter Questions You Can Ask the CEO:

- What does the next 30/60/90 days of this project look like after my trial?
- Which parts of my work would you want extended or hardened first?
- Are there parts of the codebase you would have approached differently from how I did?
- What is the long-term vision for the listing website — is it a public product or an internal tool?
- How do you measure success for this project beyond just "it runs"?
- What was the biggest red flag you saw in my work during the week?
- What was the biggest green flag?
- If I were to continue, which module would you want me to own end-to-end?
- What is the single most important habit you want from someone working with you long-term?

6 Your Final Review (Self-Assessment)

You will give yourself an honest score on each of the criteria below. I will give my own score immediately after, and we will discuss the gap. The point is not to be harsh on yourself — it is to see whether your self-perception matches reality.

Self-Review Criterion	What You Will Score Yourself On
Communication	Did you proactively update the CEO during the week? Were your blocker messages specific and actionable, or vague? How quickly did you respond?
First Meeting — Preparation & Questions	Did you arrive at the first meeting having watched all videos and read the doc? Were your questions sharp, or general? Did you leave with full clarity?
Attention to Details	Did you follow exact table names, file names (`diagnosis.md`, `failure_log.txt`, `sync_audit_log`), and the spec exactly — or did you improvise?
Deliverables & To-Do List (/50)	Score out of 50 across all to-do items in Module 1 (6 tasks) and Module 2 (3 tasks) plus the 5 deliverables. Mark each item as fully done, partially done, or missing.

Self-Review Criterion	What You Will Score Yourself On
Delivery Submission — How Clean	Was your submission organised in one folder with clear file names? Was the Loom video coherent and watchable? Were screenshots labelled?
Final Marking Discussion	You and the CEO compare scores, discuss the gap, and agree on a final mark. This determines whether the trial converts into ongoing work.

⚠ **SCOPE REMINDER:** Even in the final meeting, do not pitch new features or improvements to other modules. Stay focused on what was assigned. Discussion of next steps comes after the marking is complete.

Working With the CEO — What You Are Being Judged On

You are working directly with the CEO on this task. This is a one-week review. Here are the five things being evaluated throughout the week. Read these carefully — they matter as much as the code you deliver.

Criterion	What It Means
Communication	Keep the CEO updated proactively — do not wait to be asked. When you hit a blocker, explain clearly what you tried, what happened, and what you need, instead of sending vague messages like "it's not working."
Meeting Comprehension	Listen carefully in meetings and confirm you understand before starting work. If something is unclear, ask immediately — do not leave a meeting with unanswered questions and guess later.
Attention to Details	Follow the exact to-do list and deliverables in this document — do not skip, reorder, or add steps. Small details like correct table names, proper data formats, and clean output quality are all being watched.
Todos	Treat every item in the Module 1 and Module 2 to-do lists as non-negotiable — finish each one before marking it done. Track your own progress daily and flag any item you cannot complete as early as possible, not on the deadline.
Delivery	Submit all deliverables on time, fully working, and recorded in a clear Loom video. Last-minute submissions, missing deliverables, or broken demos will be treated as failed delivery regardless of how much work was done behind the scenes.

Come prepared. · Ask questions early. · Deliver quality work.